# Tracing Small Circuits in an Open Language Model

A Reproducibility Study of Attribution Graphs in Qwen3-4B-Instruct

MPhil in Data Intensive Science

Project 28: Mechanistic Interpretability of Open Source LLMs


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**Abstract**

Anthropic's *On the Biology of a Large Language Model* uses sparse features, attribution graphs and causal interventions to formulate hypotheses about mechanisms underlying language-model behaviour. The present study evaluates the transferability of that methodology to the open-weight, 4-billion-parameter `Qwen3-4B-Instruct-2507`. Two behaviours were studied: addition, focusing on propagation of a carry between decimal columns, and factual recall, instantiated by SI-unit and capital-city prompts. An independent pipeline generated synthetic prompt sets, captured final-position MLP outputs at seven layers, trained one 8,192-latent sparse autoencoder (SAE) per layer, constructed gradient-based contrast attribution graphs, and performed feature inhibition, latent swaps, raw-activation patches and component knockouts.

The results provided partial support for methodological transfer. The clearest factual-recall effect was a full-latent force-to-energy patch: on the energy prompt, the probability of the force prefix `new` rose from 0.0240 to 0.4980 and became the top prediction, while the energy prefix `j` fell from 0.7969 to 0.2354. Graph-selected sparse swaps were much weaker. In arithmetic, progressively ablating 73 graph-positive features reduced the correct-minus-dropped-carry logit gap only from 7.12 to 6.63. A stronger full-latent patch reduced it to 0.74, whereas a raw MLP patch transferred the source answer digit completely. A control source whose answer digit was 2 transferred 2 rather than a generic no-carry state, indicating that the intervention did not isolate carry as an independent variable. Capital-city swaps remained weak even for high-confidence prompts.

The experimental workflow was therefore reproduced, and causal structure was identified at the level of broad activations, but no Anthropic-like compact sparse circuit was recovered. Several methodological factors plausibly account for this discrepancy: the present SAEs were independently trained on only 800 final-position vectors per layer, their unconstrained $\ell_1$ objective has a scale degeneracy, and several all-position interventions applied them outside their training distribution. The findings consequently identify which components of circuit tracing transferred under a lightweight open-model implementation and which did not.

# Contents

# 1 Introduction

Large language models can produce correct answers without exposing the internal computations that produced them. Mechanistic interpretability attempts to replace a purely input–output description with a causal account in terms of internal representations and interactions. This is difficult because a model neuron need not correspond to one human-understandable concept: multiple features may be represented in superposition, making individual neurons polysemantic [5]. Sparse autoencoders offer one possible change of basis. They reconstruct an activation with a relatively small set of active learned directions, which can sometimes be more interpretable and causally useful than the original neurons [3, 4].

Lindsey et al. applied this general programme at much greater scale in *On the Biology of a Large Language Model* [1]. Their case studies use attribution graphs to propose computational pathways for behaviours including addition and factual recall, then test those pathways with interventions. The companion methods work is explicit that a graph describes a local replacement model and is a source of hypotheses, not automatically a faithful explanation of the underlying model [2]. This qualification makes the work well suited to a reproducibility study: successful reproduction requires both a plausible graph and the predicted causal effects in the original model.

The present study evaluates that transfer in `Qwen3-4B-Instruct-2507`, an open-weight causal language model with 4.0 billion parameters, 36 transformer layers and hidden width 2,560 [6, 7]. The experimental design uses small synthetic prompt corpora, independently trained SAEs at selected layers, a pruned attribution-style graph, and causal validation by inhibition and activation swapping. It thus evaluates the methodological sequence at a substantially smaller scale than Anthropic's study rather than attempting an architectural replication of the cross-layer transcoder. Addition and factual recall constitute the two principal behaviours. Factual recall is evaluated in two domains: capital cities, which parallel the target study's factual-recall examples, and SI units, which provide a science-focused extension. The SI-unit domain also permits controlled minimal pairs in which, for example, `force` is replaced by `energy` while the remainder of the prompt is held constant.

The investigation asks three questions:

1. Can a lightweight, per-layer SAE pipeline identify sparse feature sets whose interventions predictably alter the model's output distribution?
2. When sparse feature interventions are insufficient, is causal structure detectable at broader levels of the activation representation?
3. Which methodological differences explain agreement or disagreement with the Anthropic result?

Reproduction was assessed against criteria stricter than the observation of an arbitrary logit change. The clean model was required to express the behaviour; a graph-derived intervention was then required to move a pre-specified target-versus-contrast quantity in the predicted direction; and the effect was required to exceed changes attributable to numerical precision and, where possible, to survive a reverse-direction or alternative-source control. Under these criteria, the evidence supports a partial reproduction. Broad latent and raw-activation patches demonstrate causal information, whereas sparse graph features rarely control the predicted answer. The analysis therefore examines why reconstruction quality, sparse attribution and mechanistic faithfulness diverge in the present implementation.

# 2 Background and relation to the target study

## 2.1 Sparse decompositions

Let $x \in \mathbb{R}^d$ be an activation from a transformer component. A one-hidden-layer SAE encodes $x$ into an overcomplete latent vector $z \in \mathbb{R}^m$ and reconstructs it as $\hat{x}$:

$$z = \mathrm{ReLU}(W_\mathrm{e}(x - b_\mathrm{d}) + b_\mathrm{e})\,, \tag{1}$$

$$\hat{x} = W_\mathrm{d} z + b_\mathrm{d}. \tag{2}$$

When $m > d$, sparsity rather than dimensional compression is intended to make the representation useful. If only a few coordinates of $z$ are non-zero for each example, the decoder columns can be treated as candidate feature directions. Previous work has shown that such decompositions can yield more interpretable units and support causal edits in some settings [3, 4]. Neither low reconstruction error nor an overcomplete dictionary alone, however, establishes that a latent is monosemantic or causally complete.

## 2.2 Anthropic's attribution graphs

The target study did not use independent per-layer SAEs as direct substitutes for its model's computation. It used a 30-million-feature cross-layer transcoder (CLT) for Claude 3.5 Haiku [1]. A feature at layer $\ell$ reads the residual stream at that layer and has separate decoder contributions to the MLP outputs of layer $\ell$ and later layers. In simplified notation,

$$a^\ell = \mathrm{JumpReLU}(W_\mathrm{enc}^\ell x^\ell), \tag{3}$$

$$\hat{y}^\ell = \sum_{k \leq \ell} W_\mathrm{dec}^{k \to \ell} a^k, \tag{4}$$

where $x^\ell$ is the residual-stream input and $y^\ell$ is an MLP output [2]. Features across layers are trained jointly. For a particular prompt, Anthropic then constructs a local replacement model that adds the observed reconstruction error, freezes attention patterns and normalisation denominators, and exactly matches the original forward pass at the unperturbed point. Attribution graphs are pruned views of this local model. Interventions in the original model are required because a locally exact replacement can still respond incorrectly away from that point.

The present study is therefore a small-scale reproduction of the workflow rather than an architectural replication of the CLT. Table 1 summarises the distinction. This distinction constrains the interpretation of negative results: failure of the present feature basis does not imply failure of sparse-feature methods generally, while the success of a broad patch does not establish recovery of a sparse circuit.

# 3 Methods

## 3.1 Model, execution and reproducibility

All experiments used `Qwen3-4B-Instruct-2507` in non-thinking mode. The model was loaded with bfloat16 weights through Hugging Face Transformers and evaluated without sampling. Raw completion prompts were tokenised directly rather than wrapped in a chat template. Random seeds for Python, NumPy and PyTorch were fixed to 787; this seed controls dataset order, train–validation splits and

Table 1: Methodological correspondence to the target study.

| Component | Anthropic target study | Present study |
|---|---|---|
| Model | Claude 3.5 Haiku and a smaller research model | Open-weight `Qwen3-4B-Instruct-2507` |
| Dictionary | Joint cross-layer transcoder; 30 million features for Haiku | Seven independent MLP-output SAEs; 8,192 latents each (57,344 total per behaviour) |
| Training support | Features at all model layers, trained jointly | Final-token activations from 1,000 behaviour-specific prompts at layers 4, 8, 12, 16, 20, 24 and 28 |
| Local graph | Error nodes plus frozen attention and normalisation operations | Clean-value straight-through splice; local gradients through the original model; no explicit error nodes |
| Feature nodes | Active features across token positions | SAE features only at the final prompt position |
| Validation | Constrained feature interventions and mechanistic-faithfulness evaluation | Error-preserving inhibition/swap, raw MLP patching and component knockout |

SAE initialisation. The principal analyses used layers $\{4, 8, 12, 16, 20, 24, 28\}$ from the model's 36 layers.

GPU analyses were executed in Google Colab because the intended CSD3 service was unavailable during the experimental period. This choice affected the execution environment but not the architecture of the analysis pipeline: the notebooks invoke standalone scripts under `src/`, retrieve versioned code from the public GitHub repository, and persist large activation matrices, checkpoints and output artifacts separately in Google Drive. The software environment, random seed, model configuration and behaviour-specific SAE configurations are versioned. The analyses reported here correspond to `run_gpu_units_hypertrain.ipynb`, `run_gpu_math_final.ipynb` and `run_gpu_capitals_final.ipynb`. The repository is available at `https://github.com/evey-dev/test_run`.

### 3.2 Prompt sets and baseline checks

Three deterministic synthetic corpora were generated. The addition corpus contains 1,000 two-digit sums, balanced over the 100 possible pairs of ones digits with ten random tens-digit pairs for each. If the ones digits carry, the distractor is the correct total minus ten; otherwise it is the total plus ten. The units corpus contains five quantities (temperature, mass, time, force and energy), 20 contextual objects per quantity and ten templates, giving 1,000 prompts. The capitals corpus was initialised from 189 country/state examples and expanded to 1,000 prompts by substituting non-capital cities from a fixed city list.

An earlier baseline audit, retained in JSON format, used 200 addition prompts, 200 unit prompts and the original 189 capital prompts. Addition accuracy was evaluated on a short greedy completion; units and capitals used prefix-aware completion to handle multi-token answers. These results establish baseline model competence but do not constitute final held-out estimates for the expanded 1,000-prompt corpora. The SAE and intervention analyses therefore use the clean output distribution for each specific intervention prompt as their principal baseline.

Because arithmetic answers are generated autoregressively, the carry analysis required the relevant decimal position to be isolated. The principal carry prompt was

```
Question: What is 58 + 83? Answer: 1
```

which teacher-forces the hundreds digit of 141 and asks the model for the tens digit. The correct next token is `4`; the counterfactual obtained by dropping the carry from $8+3$ is `3`. The matched no-carry source was `54 + 83 = 137`, which has the same token length and the same teacher-forced hundreds digit. An alternative source, `44 + 83 = 127`, was used to distinguish transfer of a general no-carry state from transfer of the source-specific next digit.

The unit minimal pair was

```
Fact: The official SI unit for the force/energy of a moving engine thrust is named
"
```

with only `force` versus `energy` changed. Because the answers are multi-token, reported targets are first-token prefixes: principally `new` for newtons and `j` for joules. Capital pairs were Dallas/Austin versus Oakland/Sacramento, followed by higher-confidence Zarqa/Amman versus Basra/Baghdad.

### 3.3 Activation capture and SAE training

Forward hooks captured each selected MLP's output at the final input token. Each behaviour therefore produced a $1000 \times 2560$ activation matrix per layer. A seeded 80:20 split supplied 800 training and 200 validation vectors, shared across layers. Separate SAE sets were trained for addition, units and capitals to avoid conflating domain-specific activation distributions in the final comparison.

For layer $\ell$, activations were divided by

$$s_\ell = \frac{\mathbb{E}_{\text{train}} \|x^\ell\|_2}{\sqrt{2560}}, \tag{5}$$

so that the mean normalised L2 norm was approximately $\sqrt{2560}$. The SAE had input dimension 2,560 and latent dimension 8,192. It was trained for 75 epochs with Adam, learning rate $10^{-3}$, batch size 64 and $\lambda = 3 \times 10^{-4}$. With $u = x/s_\ell$, the implemented objective was

$$\mathcal{L} = \text{mean}\big[(u - \hat{u})^2\big] + \lambda \, \text{mean}(|z|). \tag{6}$$

The checkpoint with minimum validation loss was retained rather than the final epoch. Each SAE contains 41,953,792 trainable parameters; the seven models were trained independently rather than as a single 294-million-parameter network. Consequently, "lightweight" refers to computational runtime and experimental scope, not to a small statistical parameterisation.

### 3.4 Contrast attribution graphs

During graph construction, each SAE was inserted at the final prompt position using a straight-through, clean-value splice. If $\hat{x}$ is the SAE reconstruction, the modified activation is

$$x' = x + \hat{x} - \text{stopgrad}(\hat{x}). \tag{7}$$

Thus $x' = x$ in the forward pass, while gradients from the output flow through the SAE latents. This prevents compounded reconstruction error from changing the clean prediction before attribution.

For a target token $t$ and contrast $c$, the scalar objective is

$$J = \mathrm{logit}(t) - \mathrm{logit}(c). \tag{8}$$

This contrastive formulation is required for the carry analysis: a feature is relevant if it supports `4` *relative to* `3`, rather than merely increasing both digit logits. A latent's first-order node attribution is

$$A_i = z_i \frac{\partial J}{\partial z_i}. \tag{9}$$

For adjacent selected layers, the implemented edge attribution from feature $i$ to feature $j$ is

$$E_{i \to j} = z_i \frac{\partial z_j}{\partial z_i} \frac{\partial J}{\partial z_j}. \tag{10}$$

Input-token edges use the analogous embedding dot-gradient quantity. The graph retains the 20 largest-magnitude active nodes per selected layer, input nodes and edges between retained nodes whose magnitude exceeds $10^{-6}$. Decoder directions are projected through the unembedding matrix to provide three "top-output" labels. These labels are used only as diagnostics and do not establish a feature's semantic interpretation. JSON, Mermaid and interactive HTML representations are retained for each graph.

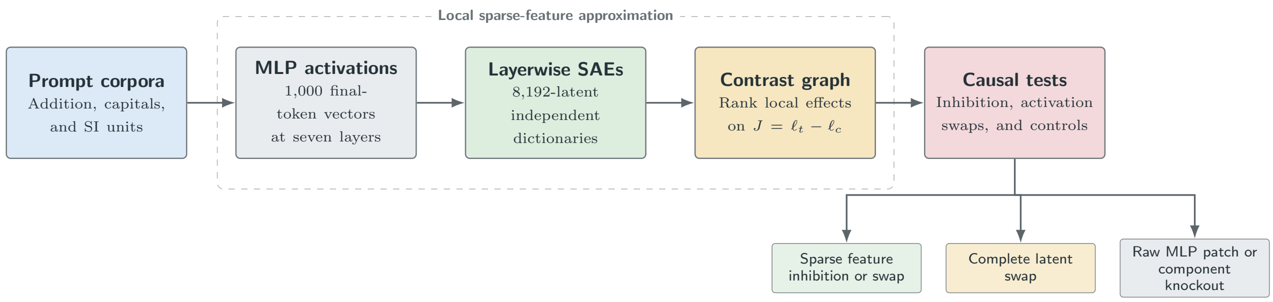


Figure 1: Independent small-scale reproduction pipeline, showing the stage at which each methodological approximation is introduced.

### 3.5 Causal interventions and evaluation

Feature inhibition is error-preserving. For a selected coordinate set $S$, let $z^{(-S)}$ equal $z$ with those coordinates set to zero. The true MLP output is edited by only the decoded feature difference:

$$x' = x + s_\ell W_{\mathrm{d}} \left( z^{(-S)} - z \right). \tag{11}$$

With $S = \varnothing$, the model is exactly unchanged. A latent swap from source $s$ to target $t$ similarly applies

$$x'_t = x_t + \alpha s_\ell W_{\mathrm{d}} (z_s - z_t), \tag{12}$$

where $\alpha$ is an edit strength. Sparse swaps replace only selected coordinates; full-latent swaps use all 8,192. Raw MLP patching replaces the selected target MLP outputs directly with source outputs and therefore bypasses the SAE. Full knockouts set an MLP, attention or transformer-block output to zero at specified token positions.

Interventions were initially applied at the final position and subsequently at all aligned positions when diagnostics indicated that final-position editing was insufficient. Token IDs and activation norms were recorded to verify that the intended non-zero locations were modified. Effects are reported using token probability, token logit and the contrast gap in Equation 8. A change in the top-1 prediction provides direct evidence of behavioural control but was not a necessary criterion: the clean arithmetic distribution is highly saturated, so a substantial logit-gap change can remain informative without altering the highest-probability token. Conversely, a prediction change under a broad, destructive intervention does not independently identify a specific circuit.

# 4 Results

## 4.1 Baseline competence

Table 2 reports the retained initial baseline audit. Every tested addition prompt was answered correctly. Performance on units and capitals was lower, partly because these answers are multi-token and because some templates elicited continuations such as `after` rather than an immediate answer. Intervention effects are therefore interpreted relative to the complete clean output distribution for the relevant prompt; low-confidence capital examples are not considered directly comparable to high-confidence arithmetic examples.

Table 2: Initial baseline audit. "Top-$k$" means that an acceptable answer was found in the saved top-20/prefix-aware evaluation. These prompt counts predate the expanded 1,000-prompt SAE corpora.

| Behaviour | Prompts | Top-1 correct | In top-$k$ |
|---|---|---|---|
| Addition | 200 | 200 (100.0%) | 200 (100.0%) |
| SI units | 200 | 134 (67.0%) | 193 (96.5%) |
| Capitals | 189 | 94 (49.7%) | 158 (83.6%) |

## 4.2 Factual recall: SI units

The clean force prompt predicted `new` with probability 0.8164 and logit 20.62. The matched energy prompt predicted `j` with probability 0.7969 and logit 19.88; `new` still had probability 0.0240 and logit 16.38. The force and energy contrast graphs each contained 20 retained feature nodes per layer. Filtering by positive node attribution selected 76 force-supporting and 97 energy-supporting features respectively.

The largest factual-recall effect was produced by the all-position full-latent swap from force into energy. It changed the top prediction from `j` to `new`: `new` rose from probability 0.0240 (logit 16.38) to 0.4980 (logit 18.38), while `j` fell from 0.7969 (19.88) to 0.2354 (17.62). The reverse energy-to-force patch did not change the highest-probability answer, but it raised the `j` logit from 10.06 to 14.81 and probability from below the displayed four-decimal resolution to 0.0026, while reducing `new` probability from 0.8164 to 0.6836. Layer-wise full-latent scans placed the largest individual movements at layers 24 and 28. For example, force-to-energy patching at layer 24 lowered the `j` logit by 1.12 and raised `new` by 1.25.

Sparse interventions did not produce comparable transfer. Swapping all 97 positive energy-graph features into force left the `j` logit at 10.06 and increased `new` from 20.62 to 20.75. Swapping 76 positive force-graph features into energy raised `new` only from logit 16.38 to 16.75 and probability 0.0240 to 0.0352; `j` remained the top token. Inhibiting the same force features at the final position reduced `new` probability from 0.8164 to 0.8008 and its logit by 0.24. Diagnostics confirmed that the targeted latents were active; the small effect was therefore not attributable to the selection of inactive coordinates.

Table 3: Main SI-unit swap results. Probabilities are for first-token answer prefixes, not whole words.

| Target run | Intervention | $p(\texttt{j})$ | $p(\texttt{new})$ | $\ell(\texttt{j})$ | $\ell(\texttt{new})$ |
|---|---|---|---|---|---|
| Energy | clean | 0.7969 | 0.0240 | 19.88 | 16.38 |
| Energy | full force latent | 0.2354 | 0.4980 | 17.62 | 18.38 |
| Energy | sparse force features | 0.8008 | 0.0352 | 19.88 | 16.75 |
| Force | clean | $< 0.0001$ | 0.8164 | 10.06 | 20.62 |
| Force | full energy latent | 0.0026 | 0.6836 | 14.81 | 20.38 |
| Force | sparse energy features | $< 0.0001$ | 0.8242 | 10.06 | 20.75 |

These results establish that the learned latent spaces contain information that can causally alter unit-token selection, particularly in the force-to-energy direction. They do not, however, establish a sparse-circuit reproduction: the observed effect occurred only when the complete latent code across seven layers and all token positions was transferred, whereas the graph-selected positive subset was insufficient.

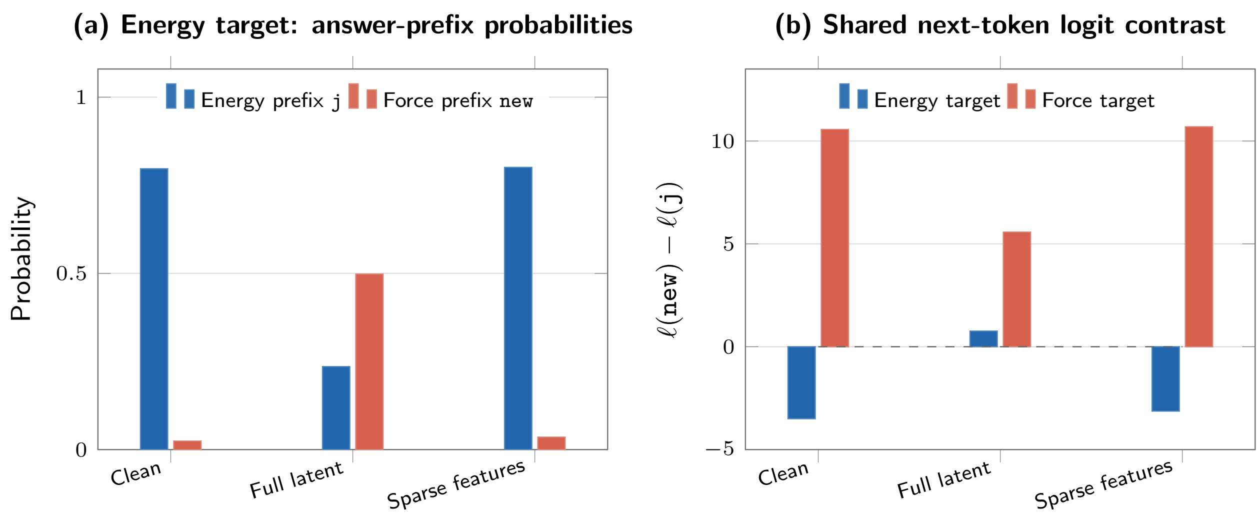


Figure 2: Full-latent unit swaps transfer answer-prefix preference, whereas the graph-selected sparse subsets do not. In panel (b), positive values of $\ell(\texttt{new}) - \ell(\texttt{j})$ favour the force prefix.

### 4.3 Addition: a carry between decimal columns

On `58+83` after teacher-forcing `1`, the clean model predicted `4` with displayed probability 1.0000 and logit 30.62. The dropped-carry token `3` had probability 0.0008 and logit 23.50, so the clean contrast gap was $J = 7.12$. On the no-carry minimal pair `54+83`, the next token `3` had displayed probability 1.0000 and logit 29.38; `4` had probability 0.0002 and logit 20.88.

Progressive inhibition of carry-graph features produced a small, non-monotonic response. At 10 features, the `3` logit increased by 0.12 while `4` was unchanged. At 50 features, `4` decreased by 0.12 and `3` increased by 0.25. With all 73 positive-attribution features, `4` decreased by 0.38 and `3` increased by 0.12, reducing the gap from 7.12 to 6.63. The direction is consistent with the contrast graph but the effect is too small to establish a compact carry circuit.

Full-latent swaps were stronger. At edit strength 1, patching `54+83` into `58+83` increased the `3` logit from 23.50 to 24.12 and probability from 0.0008 to 0.0017; `4` remained at probability 0.9961. At strength 2, `3` rose to probability 0.3203 and logit 27.88, while `4` fell to 0.6797 and 28.62. The contrast gap decreased to 0.74. By comparison, swapping only the 80 positive features from the no-carry graph did not produce the predicted directional effect: the `4` logit increased from 30.62 to 30.75 and `3` remained at 23.50.

Raw all-position MLP patching bypassed the SAE and produced a complete transfer of the source answer at the displayed precision. Patching `54+83` into `58+83` changed the top token from `4` to `3`, with `3` logit 29.62 and `4` logit 21.25. The reverse patch changed the no-carry target from `3` to `4`. These interventions establish that the selected-layer MLP outputs contain causally sufficient information to determine the next answer digit, but they do not identify the semantic content of that information.

The alternative-source control clarifies this ambiguity. `44+83=127` is also no-carry in the ones column, but its teacher-forced next digit is `2`, not `3`. A strength-2 full-latent patch into `58+83` made `2` the top token with probability 0.6523; `3` also rose to 0.3477 and `4` fell to 0.0001. The raw MLP patch transferred `2` with displayed probability 1.0000. The successful raw patch therefore primarily transfers source-specific next-digit state and does not independently establish the transfer of an abstract "carry 1" feature.

Table 4: Arithmetic interventions on the `58+83` tens-digit target. The clean logits are $\ell_4 = 30.62$ and $\ell_3 = 23.50$.

| Intervention | $p(4)$ | $p(3)$ | $\ell_4$ | $\ell_3$ | Interpretation |
|---|---|---|---|---|---|
| Clean | 1.0000 | 0.0008 | 30.62 | 23.50 | Correct, saturated |
| 73 positive features inhibited | 0.9961 | – | 30.25 | 23.62 | Small directional shift |
| Full `54+83` latent, $\alpha = 1$ | 0.9961 | 0.0017 | 30.50 | 24.12 | Partial shift |
| Full `54+83` latent, $\alpha = 2$ | 0.6797 | 0.3203 | 28.62 | 27.88 | Large but non-sparse shift |
| Raw `54+83` MLP | 0.0002 | 1.0000 | 21.25 | 29.62 | Source digit transferred |
| Full `44+83` latent, $\alpha = 2$ | 0.0001 | 0.3477 | 18.00 | 26.75 | Source digit 2 becomes top |

The all-position MLP knockout provides complementary evidence. Zeroing the MLP outputs at all seven selected layers reduced `4` probability to 0.6406 and logit to 27.25, while increasing `3` to probability 0.1836 and logit 26.00. However, `2` also rose to probability 0.1426 and logit 25.75. Layer 24 alone produced the largest disruption: the model predicted `2` with probability 1.0000, while the `4` logit fell by 10.88. Thus layer 24 is causally important for maintaining the answer, but the knockout does not show that it represents a carry-specific variable. The result instead demonstrates a broader dependence of digit selection on this layer's MLP output.

### 4.4 Factual recall: capital cities

The original Dallas/Oakland pair had low baseline confidence. Dallas predicted ` Austin` with probability 0.2773, only slightly ahead of ` Dallas` at 0.2158. For Oakland, ` Sacramento` had probability 0.2217 while ` after` was top at 0.4141. A full Oakland-to-Dallas latent swap raised the Austin probability to 0.3281 but did not transfer Sacramento, whose logit rose only from 2.67 to 3.97. In the reverse direction, Sacramento rose from 0.2217 to 0.3359, but ` after` remained top at 0.3809. Sparse swaps were smaller still.

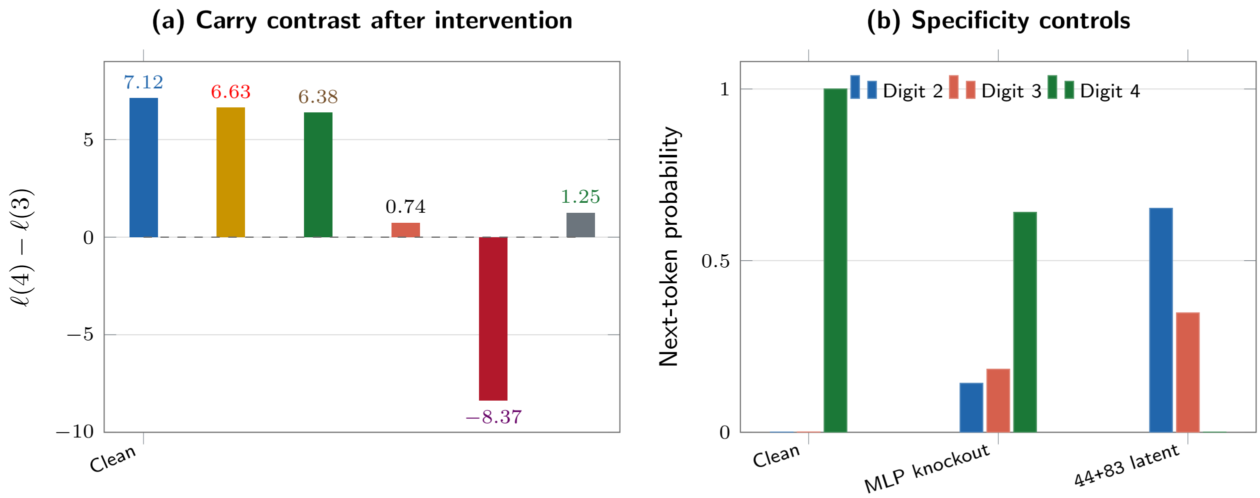


Figure 3: Arithmetic interventions reveal broad causal answer state but do not isolate a carry-specific sparse feature group.

Low baseline confidence was a plausible confound; a second pair therefore used Zarqa to Amman and Basra to Baghdad. Clean first-token probabilities were 0.9844 for `Am` and 0.9648 for `Baghdad`. Higher baseline confidence did not produce stronger causal transfer. Patching Basra into Zarqa raised the Baghdad logit from 6.97 to 12.62, but its probability remained 0.0002 and `Am` remained top at 0.9805. The reverse patch reduced Baghdad probability from 0.9648 to 0.8711 and raised the Amman prefix from 0.0001 to 0.0002. Graph-selected sparse swaps caused negligible changes in both directions. Capital recall therefore constitutes a weak or negative reproduction whose outcome cannot be attributed solely to low prompt confidence.

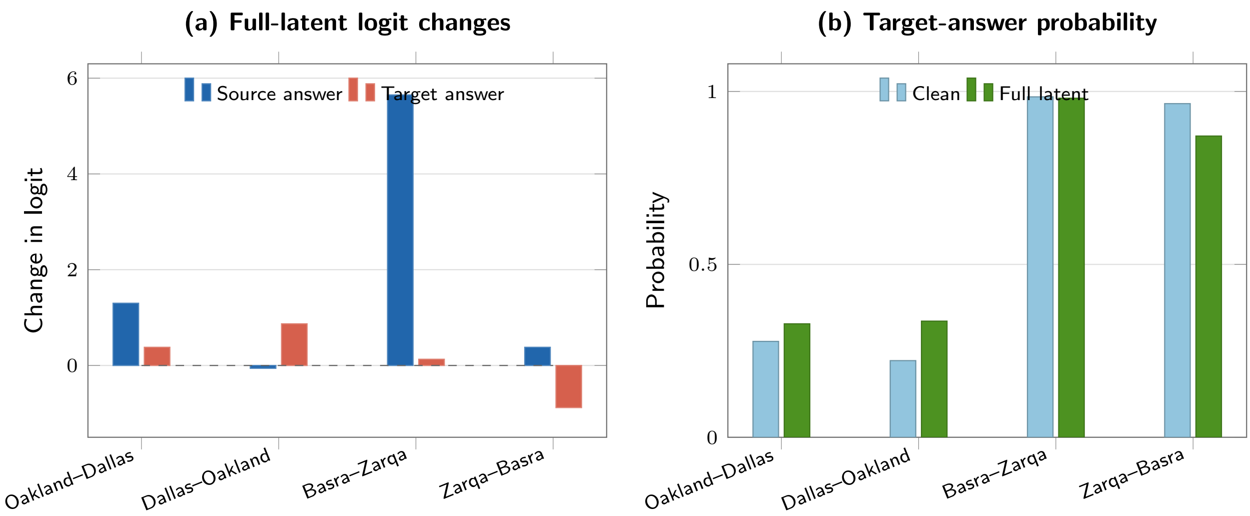


Figure 4: Capital-city full-latent interventions. Panel (a) reports changes in the source- and target-answer logits; panel (b) compares the target-answer probability before and after intervention. The high-confidence Basra–Zarqa control reduces one potential confound but does not yield clear answer transfer.

# 5 Discussion

## 5.1 Extent of reproduction

The first research question yields a qualified negative result. Aggregate inhibition of graph-selected features occasionally moved the pre-specified contrast in the expected direction, most clearly in the arithmetic scan. However, no graph-selected sparse swap produced a clear transfer of a tested answer. The evidence is therefore insufficient to classify the result as a sparse-circuit reproduction.

The evidence concerning the second research question is more affirmative. Full latent codes and raw MLP outputs contain causally effective information. The force-to-energy unit patch changes the most probable answer prefix, the stronger no-carry arithmetic latent patch nearly equalises the correct and dropped-carry digits, and raw MLP patches transfer source digits in both directions. These controls distinguish two possible failure modes: transferable internal state is present in the model, but the selected sparse coordinates do not constitute a compact causal basis for that state.

The alternative arithmetic source substantially constrains interpretation of the apparent transfer. In its absence, the `54+83` $\to$ `58+83` prediction change could be interpreted as transfer of a no-carry mechanism. The transfer of `2` from `44+83` instead supports the narrower conclusion that broad patches transfer source-specific answer state. Evidence for a carry variable would require generalisation across multiple source–target pairs in which carry status and output digit are independently varied.

The findings reproduce the methodological structure of the target study but are substantially weaker at the level of sparse causal control. The study established the behaviour, selected a decisive logit contrast, constructed a pruned feature graph, derived interventions and evaluated them in the underlying model. It did not recover a compact sparse-feature mechanism. This outcome is consistent with the target study's report that cross-layer transcoders outperform per-layer alternatives and that mechanistic faithfulness requires empirical evaluation rather than assumption [2].

## 5.2 Divergence between reconstruction and sparse causal control

Several methodological differences provide testable explanations for the gap between reconstruction performance and sparse causal control. These explanations do not require an unsupported assumption that the discrepancy arises solely from model-specific properties of Qwen.

**The implemented $\ell_1$ objective is scale-degenerate.** Equation 6 penalises latent magnitude, but the decoder columns are not constrained to unit norm. For any $c > 0$, consider the rescaling

$$W'_{\mathrm{e}} = cW_{\mathrm{e}}, \qquad b'_{\mathrm{e}} = cb_{\mathrm{e}}, \qquad W'_{\mathrm{d}} = \frac{1}{c}W_{\mathrm{d}}. \tag{13}$$

Positive homogeneity of ReLU gives $z' = cz$, while $W'_{\mathrm{d}}z' = W_{\mathrm{d}}z$. Reconstruction is unchanged but the $\ell_1$ term is multiplied by $c$ and can, in principle, be made arbitrarily small. This property does not demonstrate that optimisation fully exploited the degeneracy, and ReLU can still produce exact zeros. It does establish that the raw $\ell_1$ term is not an invariant measure of sparsity and that $\lambda$ alone cannot guarantee a sparse code. Decoder normalisation, an $\ell_0$-oriented nonlinearity such as JumpReLU, or a scale-aware penalty would remove or reduce this ambiguity. The target CLT used a decoder-norm-aware Tanh penalty rather than the present raw $\ell_1$ term [2].

**The dictionaries are underconstrained by the activation sample.** Each per-layer SAE has approximately 42 million trainable parameters but only 800 training vectors. The 1,000 prompts are also highly templated: units span five quantities and ten repeated syntactic forms. Low validation reconstruction loss on the same prompt distribution therefore does not establish feature uniqueness, monosemanticity or causal generalisation. Final $\ell_0$ firing-rate, dead-feature and feature-consistency diagnostics were not retained. Consequently, no quantitative claim about achieved sparsity is made.

**Training and intervention supports do not always match.** SAEs were trained only on final-token MLP outputs. All-position latent interventions then encoded every prompt position, including words and digits, with those final-position dictionaries. This is out of distribution by construction. The mismatch is most pronounced in arithmetic: training prompts ended at `Answer:`, whereas the carry graph and swaps operate after a teacher-forced `1`. The exact force prompt also appears in the generated unit corpus, whereas the synthetically modified energy minimal pair does not; the principal unit result therefore does not constitute a strictly held-out generalisation test. These limitations do not invalidate the observed forward-pass changes, but they restrict the extent to which those changes support claims about stable, distribution-independent features.

**The graph is a smaller local approximation.** Only final-position SAE features are nodes. Attention and normalisation are not frozen, and reconstruction discrepancies are not represented as explicit error nodes. The straight-through splice guarantees the clean value, but its gradients remain a first-order local approximation through the original nonlinear model. The graph can consequently rank locally sensitive coordinates without providing the same replacement-model semantics as Anthropic's CLT graph. Positive node attribution is also not guaranteed to imply a large finite ablation effect when features interact nonlinearly.

**Saturation and intervention specificity matter.** The clean arithmetic gap of 7.12 logits leaves little probability mass on `3`; small but real logit changes are visually compressed. Increasing edit strength exposes a larger effect, but also moves farther from the local regime in which gradient attribution is most defensible. Component knockout presents a different inferential limitation: it establishes causal dependence but is too destructive to provide mechanistic specificity. The increase in both `2` and `3`, together with the layer-24 change to `2`, demonstrates that a change in the highest-probability token does not by itself identify a carry circuit.

### 5.3 Threats to validity

The study is exploratory and uses selected prompts rather than a registered large-sample intervention benchmark. The forward passes are deterministic, but only one SAE seed was used, so feature identities and sparse effects have no between-seed stability estimate. Prompt corpora supply train and validation splits for reconstruction, but the intervention prompts were not consistently reserved as an independent test set. Aggregate baseline outputs were retained from an earlier, smaller dataset version, while final SAE training used the expanded corpora. These baseline percentages should therefore be treated as competence checks, not confidence intervals for final test accuracy.

Tokenisation also constrains interpretation. Unit and capital answers are often multi-token, so the experiments intervene on first-token prefixes rather than complete answer strings. Arithmetic teacher forcing isolates a decimal position but changes the activation context relative to SAE training. Finally, bfloat16 execution and single-run reporting make changes of roughly 0.1 logits weak evidence unless they form a coherent pattern or are supported by stronger controls.

A post-experiment code audit identified one reporting defect: the saved raw-MLP layer-scan path patched every requested layer in each nominal single-layer row. Consequently, the repeated per-layer raw-scan rows are excluded from the present analysis. The aggregate raw MLP patches, full knockouts and SAE layer scans use separate execution paths and are unaffected. The hook-selection error has been corrected in the repository. Exclusion of the affected rows prevents an invalid layer-localisation inference.

### 5.4 Prioritised future work

The highest-priority extension is to align the random variables represented in SAE training with those modified during causal evaluation. Increasing the number of training epochs without resolving this mismatch would provide limited additional evidence.

1. Capture activations at all token positions, or at a pre-specified set of intervention positions, and include teacher-forced partial answers in the arithmetic SAE corpus.
2. Retrain with decoder-column normalisation or a gated/JumpReLU objective; report $\ell_0$ firing counts, dead-feature rates, reconstruction variance and clean-model recovery before interpreting features.
3. Build a held-out factorial carry benchmark. Carry status, source next digit and target next digit should vary independently across many matched pairs. Rank features by their marginal effect on $\ell(\text{correct}) - \ell(\text{dropped carry})$ across examples rather than on a single prompt.
4. Repeat SAE training across seeds and test whether feature groups, not exact feature indices, reproduce their causal effects.
5. Subject to available computational resources, replace independent output SAEs with per-layer transcoders or a small cross-layer transcoder and add explicit residual/error nodes. Compare model-level recovery and perturbation faithfulness at matched feature budgets.
6. Extend the graph to attention or residual-stream features. The weak sparse MLP result does not establish that the missing mechanism is absent; it may be represented in attention-mediated state or SAE reconstruction error.

These steps would discriminate among the explanations advanced here. In particular, improved sparse effects after position-matched training would support distribution mismatch as the main cause; persistence of broad raw-patch effects without stable sparse features would support a genuinely distributed or interaction-dependent representation.

## 6 Conclusion

The present study independently implemented a small-scale mechanistic- interpretability pipeline for an open-weight language model. The pipeline generated and audited behavioural prompts, captured selected MLP activations, trained per-layer SAEs, constructed contrast attribution graphs, and evaluated them using feature inhibition, latent swaps and activation-level controls. The principal positive finding is a hierarchy of causal effects rather than an isolated feature: raw MLP state can transfer an arithmetic answer; complete SAE latent state can substantially alter arithmetic and unit predictions; and sparse graph-selected subsets generally cannot. Capital recall provides a negative result even after baseline confidence was increased.

The evidence therefore supports transfer of the experimental methodology more strongly than recovery of a sparse circuit. Under the lightweight Qwen implementation, attribution graphs generate testable causal hypotheses, but their highest-attribution SAE nodes do not consistently constitute a compact, faithful mechanism. The mathematical and methodological analysis identifies four specific limitations—scale-degenerate sparsity, limited activation support, positional distribution mismatch and a less expressive local replacement model—that yield testable explanations for the partial reproduction. The study thereby establishes both whether the target effects were observed and the conditions under which the corresponding mechanistic inferences are warranted.

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## Declaration of use of autogeneration tools

Generative AI tools, including OpenAI Codex, were used during software development to assist with code review, debugging, experimental documentation, and the structuring and drafting of this report. GPU experiments were executed by the candidate. The candidate remains responsible for checking the final analysis, numerical claims, citations, wording and conclusions. This declaration describes the use of these tools in the preparation of the submitted work.